



Market structure: fundamentals

SMM921: Trading and Microstructure

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$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Roadmap I

In this section we will introduce the fundamentals of how market structures look and how markets operate. Fundamentally, we will need to explain;

- Players in the trading game and their characteristics/motivations
- Rules of the trading game
- Outcomes of the trading game

Roadmap II

In more specific terms, this will lead us to characterise;

- Trader types and features
- Order types
- Basic market structures
- Basic market quality measures: liquidity and efficiency

How are securities traded on exchanges: traders and order types I

Fundamental question about traders?

- How pressing are your trading needs i.e. how impatient are you?
- Equivalently, how much are you willing to pay to get swift execution?

The answer to this question will affect your trading strategy and order choice. It will depend on why you are trading;

- A trader using accounting information to predict long run returns will likely not care much whether they trade today or next week.
- A trader who uses signals derived from high-frequency price movements will likely want to trade on them within the next few milliseconds.

How are securities traded on exchanges: traders and order types II

How are market quotes generated and how do trades come about?

Two basic **order types**;

- **Market order**; an order to buy/sell stock to be executed immediately at the best price available.
 - Market order traders **pay** for immediate execution
 - Market orders give you **execution certainty**

buy now no matter the price then gg
dont care about the price i want it now
YAY

pros trade now
cons pay and price uncertainty

How are securities traded on exchanges: traders and order types II

- **Limit order**; an order to buy/sell stock that only executes if a specified price is reached.
 - Example: a limit buy might specify a maximal price of £15 and will only execute if a counter-party can be found who wishes to sell at £15 or below.
 - Limit orders traders **get paid** for being patient. opposite
less cost but execution uncertainty
 - Limit orders are not certain to execute.
 - [Exception: a **marketable limit order** is one for which the price limit is such that it will execute immediately e.g. if there is an existing limit sell with a price of 100, a limit buy order with price 105 will immediately execute. It is effectively a market order]

The vast majority of order submissions to modern equity markets are limit orders and most of them are cancelled/withdrawn rather than executed.

Market structure aside: transparency I

Transparency: are orders to buy/sell and the resulting trades observable by the broad set of market participants?

- Over the long term, equity markets have tended to become more transparent.
 - First trades and then quotes and then all orders have become visible in many venues
- Recently, though, class of opaque trading venues has emerged: **dark pools** the point is the secrecy
- Dark pools co-exist alongside transparent trading venues
- In the last 10 yrs, regulatory push to make derivatives markets more transparent.
 - Helps customers see what the 'right price' is at a point in time.
 - Might increase trading activity and effectiveness if trading process more visible
 - But might help a strategic counterparty **front-run** your order.

Market structure aside: transparency II

- **New development:** consolidated tape in the EU and UK
 - Both EU and UK are now establishing consolidated tape providers.
 - These are centralised, real-time data feeds aggregating trade and quote information across all venues.
 - Currently, fragmented data forces brokers to build expensive proprietary aggregation infrastructure, the tape solves this
 - EU bond tape expected from late 2025; equity tapes to follow in both EU and UK
 - Significant step toward genuine post-trade transparency in fragmented markets

consolidated tape for equity
taking all the fragmented information from around the world into one data feed
undo the fragmentation

How are securities traded on exchanges: market types I

Auction versus dealer markets: the two key different types of market organization.

- **Auction market:**

brokers in between but directly trade one another
price is set by SD

ask sell
bid buy

- Bid and ask prices are set by public limit orders.
- The bid is the highest outstanding limit buy price and the ask is the lowest unexecuted limit sell price.

- **Dealer market:**

he sets the price and if you want you go with him
he rebalances his portfolio changing bid and ask to secure his liquidity

- Designated market-makers post their own bid-ask quote pairs
- The market bid/ask comprises the most competitive of the sets of bids and offers.

- **Hybrids:** most real-world trading structures mix features from auction and dealer markets. Very few pure examples of either exist.

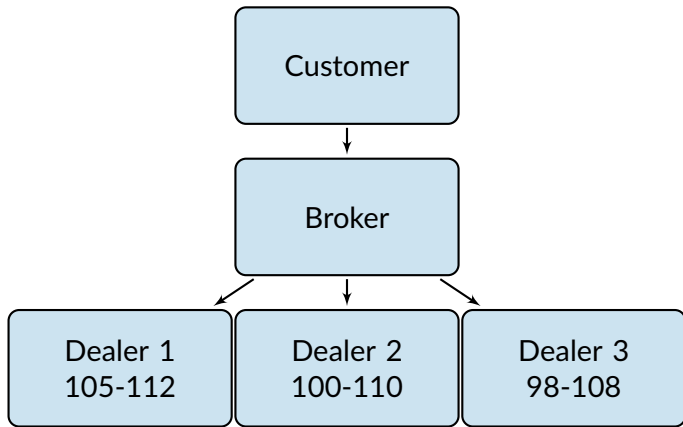
most venues do both
the dealer offering a price and the rest is
done through SD

How are securities traded on exchanges: market types II

Thus:

- In an auction market, investors trade directly with other investors.
- In a dealership, all investor trades are executed against a dealer.
- In a hybrid, there's a bit of both.

Market structure: dealership I



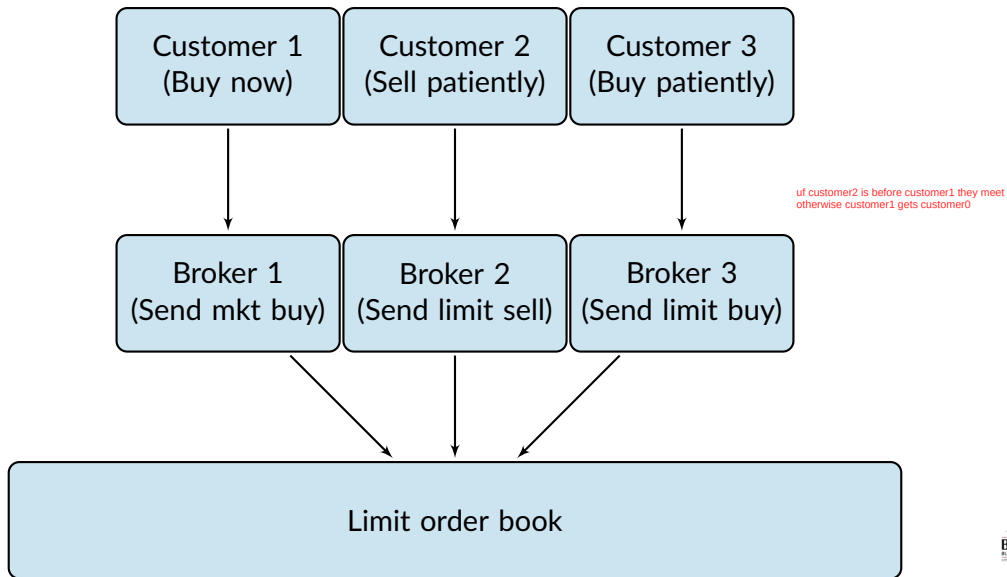
Market structure: dealership II

Features:

- Customer communicates trade to broker
- Broker takes trade to market
 - Surveys dealers and their prices (electronically or over the phone or physically)
 - Dealers compete on price, but may charge different prices depending on inventory/information
- In the example;
 - Best bid-ask is 105-108
 - Customer buy order will go to dealer 3
 - Customer sell order will go to dealer 1
 - Dealer 2 is entirely uncompetitive.

Where/when do we see this structure? Equity markets before the 1990s (LSE and Nasdaq looked like this) and many **OTC markets** for derivatives today.

Market structure: centralized limit order book I



Market structure: centralized limit order book II

Comments:

- Customer tells broker direction of trade (i.e. buy/sell) and how patient (s)he is;
 - Impatient traders submit market(able) orders (e.g. buy now regardless of price)
 - Patient investors submit limit orders (e.g. sell at a price no less than 110)
 - Market orders always execute but pay for their impatience.
 - Limit orders have uncertain execution but, if they execute, benefit from being patient
- Central limit order book (CLOB) collates orders and publishes best bid and offer (BBO)
 - Rank orders in terms of price/time/size
 - Best bid is limit buy with highest price
 - Best offer is limit sell with lowest price
 - Bid-ask spread is the difference between the best offer and the best bid
 - Incoming market buy (sell) orders execute against best offer (bid)
- CLOB may also publish details (prices and sizes) of all limit orders outside the best

Market structure: systematic internalisers I

What is a systematic internaliser?

bringing trading into your own balance
sheet but need to publish orders

- An investment firm, typically a large bank, that executes client orders by dealing from its own account, rather than routing them to a trading venue
- Trading is bilateral: you trade against the bank's book directly, not against other investors
- SIs must publish firm quotes under regulatory rules (MiFID II/MiFIR in Europe), so they are not opaque.
- Have become a third major venue type, alongside lit markets and dark venues

Market structure: systematic internalisers II

How SIs differ from other venue types:

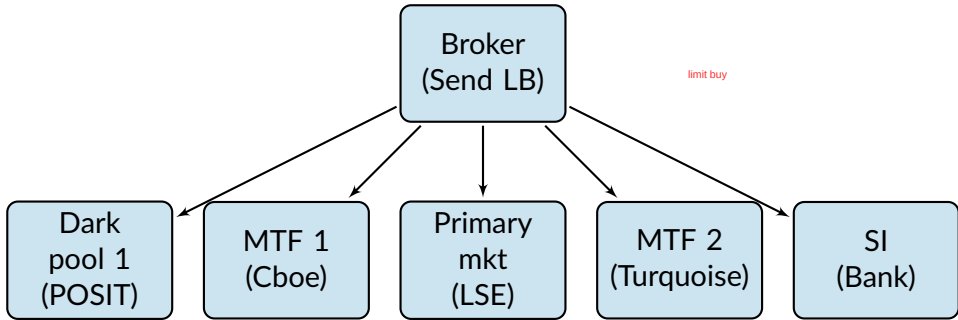
- vs. CLOB: no anonymous order book, one is trading with a single named counterparty
- vs. Dark pool: SIs have pre-trade quoting obligations; dark pools have none
- vs. pure OTC: SIs operate within a defined regulatory framework with transparency requirements

Why SIs matter

- Have grown substantially since MiFID II came into force (2018)
- Along with other off-exchange mechanisms, SIs mean that primary exchanges now account for less than a third of total European equity trading volume
- Represent a fundamentally different model: the broker who routes your order and the liquidity provider who fills it can be the same institution

Market structure: modern fragmented equity markets I

Focus on a particular broker who has received a patient buy order from a particular customer



Market structure: modern fragmented equity markets II

Comments:

- **Basic feature: fragmentation.** brokers face genuine choice regarding where to route any order, and order flow thus splits across many venues.
- In the EU upwards of 55 venues
- Primary exchanges account for less than 1/3 of total European equity trading value.
- Main venue type choices;
 - **Primary market** e.g. the LSE, usually a CLOB
 - **Multilateral trading facilities** (MTFs) e.g. Cboe Europe – structured similarly to primary venues, competing on cost and technology.
 - **Dark pool** – electronic LOB, with prices constrained by the primary, but no pre-trade transparency (e.g. POSIT)
 - **Systematic Internaliser** (SI): a bank executing orders against its own book, with requirement to publish quotes.
 - **Periodic Auction venue:** transparent batched auctions running regularly throughout trading day (reduces the advantage of the quickest traders).

Market structure: modern fragmented equity markets III

Implications for execution strategy;

- Could split the client's order and send portions to multiple trading venues simultaneously
- Could use a **smart order router** (SOR) to decide which venue(s) to send the order to
 - The router aggregates data from across venues and computes a consolidated view of prices and available size
 - Data aggregation problem: without a consolidated tape, brokers must build proprietary data infrastructure to see where liquidity sits. This technology cost favours large institutions
 - Forthcoming change: both the EU and UK are establishing consolidated tapes, expected to reduce this data asymmetry and make best execution more achievable for all market participants

a consolidation tape is a way to solve the same problem that the SOR is trying to solve

Market structure aside: payment for order flow (PFOF) I

Question for a retail trader: in a fragmented world, does my broker actually send my order to the best venue for you?

What is PFOF?

- A market maker or trading venue pays a broker a fee in exchange for the broker routing client orders to them
- The market maker profits from the bid-ask spread and rebates a portion to the routing broker
- The broker faces a conflict of interest: they may route your order to whoever pays them most, rather than whoever gives you the best price

Market structure aside: payment for order flow (PFOF) II

The economics:

- Zero-commission retail trading (Robinhood, Trade Republic, etc.) is largely funded by PFOF and, so, the cost to the investor is shifted from explicit commissions to potentially inferior execution quality
- In the US, three wholesalers handle over 80% of retail equity orders; US brokerages collectively earned \$3.8 billion in PFOF revenue in 2021 alone

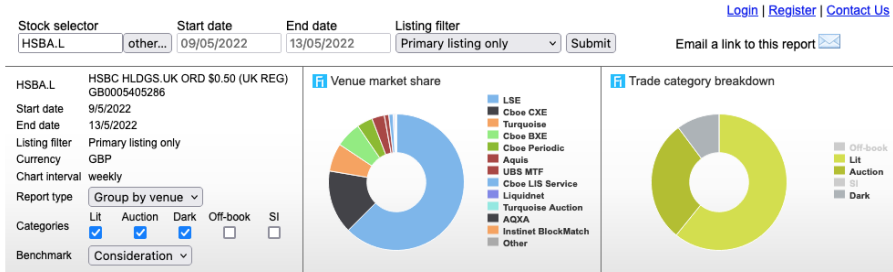
A striking regulatory contrast:

- PFOF banned in the UK since 2012 (reinforced in 2018)
- EU banned PFOF: becoming effective in June this year
- In the US, SEC proposals to restrict PFOF were withdrawn in 2025

PFOF will recur in Lecture 3 when we examine the GameStop episode and the retail trading revolution.

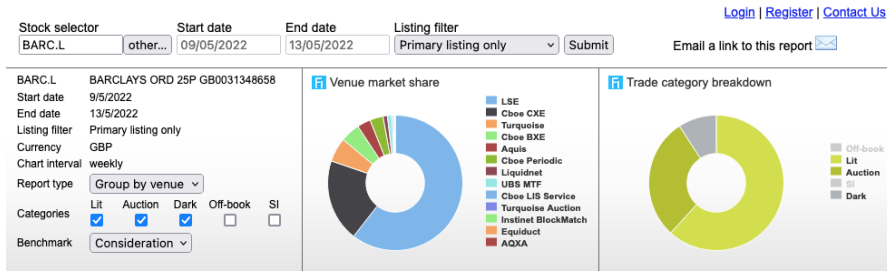
Fragmentation: HSBA

fragulator
now more opaque sad



since 2022 the distribution of HSBC shares
of LSE has been cut in half

Fragmentation: BARC



Fragmentation: the issues I

For the economist:

- **Competition:** between trading venues might be regarded as a good thing;
 - Drive down explicit costs of trading on venues
 - Supply a variety of trading mechanisms so as to suit heterogeneous trader types
- **Benefits of scale:**
 - Economies of scale in providing trading services.
 - Trading efficiency is fundamentally about finding counter-parties – so it makes sense to cluster counter-parties and order flow into one place to exploit **network externalities**

Fragmentation: the issues II

- **Liquidity measurement:**

- Is total market depth just the sum of depths across venues?
- What happens if traders post limit orders in more than one venue? (The "phantom liquidity" problem – the same liquidity may appear simultaneously on multiple venues but can only be consumed once)

- **Market quality:**

- Does fragmentation impede price discovery?
- What is the impact of fragmentation on final trading costs for investors?
- Recent evidence: fragmentation reduces average spreads in normal times but may make liquidity more fragile in stress episodes, so that liquidity deteriorates sharply when it is most needed (more on this in Lecture 2)

Fragmentation: the issues III

For the trader:

- **How to ensure one gets best execution?**
 - Must monitor all relevant venues, generating a significant technology cost
 - Smart order routers help, but are only as good as the underlying data
 - Consolidated tapes partially address the data aggregation problem
- **Are some venues more 'toxic' than others?**
 - Venue toxicity: the degree to which trades on a given venue tend to be executed against informed counterparties who trade against you
 - Dark pools: designed to reduce toxicity for large traders, but evidence is mixed.
 - SIs: banks typically internalise uninformed (retail) flow, keeping it off the lit market. This benefits the retail clients, but reduces the pool of natural counterparties on lit venues
- **PFOF and routing conflicts:**
 - In markets where PFOF operates (notably the US), the broker's routing decision may not align with the client's best execution interest
 - Regulatory response varies sharply by jurisdiction
- **Liquidity measurement:** same problem as for academic.

Fragmentation: some evidence I

Foucault and Menkveld (2008); [▶ Published paper](#)

- Look at competition for trading in Dutch stocks (EuroSETS, London, versus Amsterdam)
- Depth is greater in more fragmented markets

US evidence from O'Hara and Ye (2011) [▶ Published paper](#)

- Match more fragmented with less fragmented stocks and compare trading.
- Fragmentation reduces spreads and execution times for trades in large stocks.
- Spreads also smaller for small stocks.
- Some evidence that return volatility higher under fragmentation but also prices more efficient.

Fragmentation: some evidence II

The stress-period problem:

- BIS Working Paper (2024): ▶ Working paper
 - Across a large cross-country panel, both algorithmic trading and fragmentation are associated with lower average bid-ask spreads
 - Consistent with competition improving liquidity in normal times. However, both are also associated with higher skewness of spreads, meaning liquidity becomes sharply worse precisely during market stress episodes

The emerging consensus: fragmentation reduces costs on average, but may generate liquidity problems precisely when markets need liquidity most (i.e. in stress events).

Liquidity

In this section we will discuss the following;

- What is market liquidity?
- Why does it matter?
- How is it typically measured?

Definitions: Liquidity I

In a **liquid** market, costs of trading are likely to be small. A liquid market is often defined as one in which one can trade a large block of stock;

- Quickly (i.e. no delay or missed trading opportunities)
- Cheaply (i.e. small spreads)
- Without moving the price (i.e. small market impact)

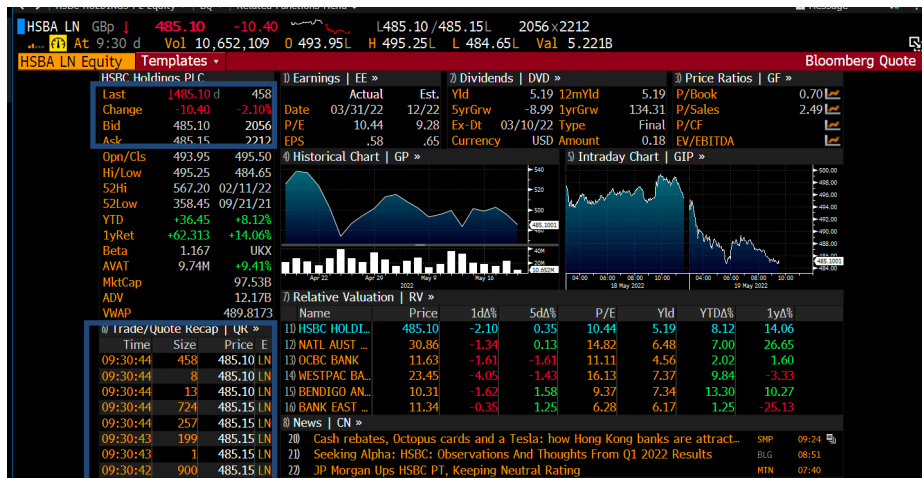
Definitions: Liquidity II

So: in a liquid set of markets one can;

- Swiftly modify one's portfolio in response to new information
- And do so at low cost so that one's portfolio returns are not substantially eroded.

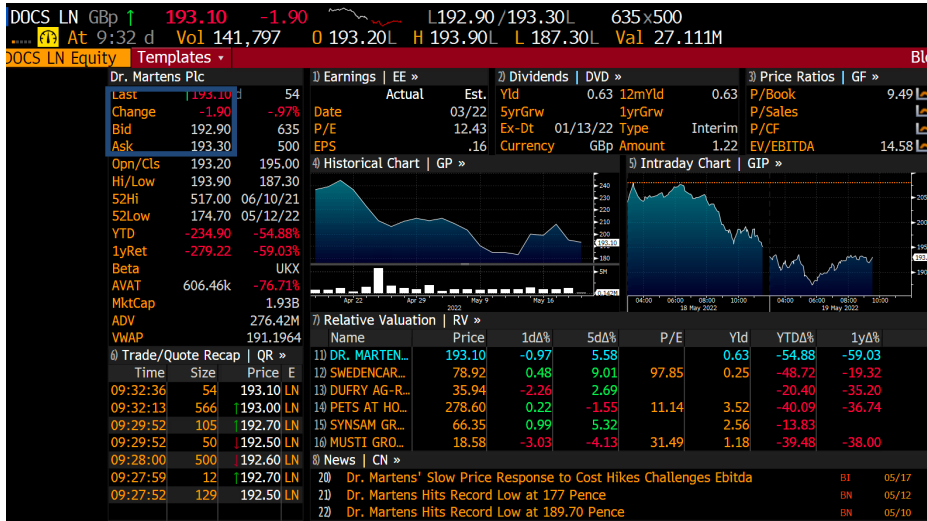
One concern: markets often become illiquid precisely when one wants to trade (e.g. in crashes and panics).

Market quotes: example HSBA



the last orders in the last 2 secs

Market quotes: example DOCS



Liquidity measurement

How you measure liquidity will depend on your style/size of trading. Compare;

- A **retail trader** who modifies their portfolio once a quarter and is a tiny fraction of market volume.
- A **quant hedge fund** who trades every day, throughout the day and potentially trades up to 1% of the average day's volume.

There will/should be feedback between liquidity and trading style, though;

- **Measurement:** how you trade will affect the liquidity metrics that are relevant to you
- **Trading style:** the level of liquidity in a particular asset will affect how you trade it.

Spreads I

Denote the market bid price for a stock by P_b and the ask/offer price by P_a .

- The **bid** is the price that one can immediately sell a unit of stock.
- The **ask** is the price at which one can immediately buy a unit.

From these we can derive;

- **Midquote**: the simple average of the bid and ask.

$$\text{Midquote} = \frac{1}{2}(P_b + P_a)$$

Think of this as market view of fair value of the stock.

both people tugging at the rope

Spreads II

- **Percentage Bid-Ask spread:** is the distance between bid and offer expressed as a percentage of the midquote;

$$\text{Spread} = 10,000 \times \frac{\overset{\text{in bips}}{P_a - P_b}}{\overset{(\text{ask} - \text{bid}) / \text{midquote}}{\text{Midquote}}}$$

The 10,000 factor gives a measure in basis points (i.e. hundredths of a percentage point)

Spreads III

Spread interpretation: if the midquote is fair value, (half of) the spread is a measure of trading costs – telling you how much above fair value the purchase price is and how much below fair value the sale price is.

Example: for the previous DOCS data;

$$\text{Midquote} = 193.1 \quad , \quad \text{Spread} = 10,000 \times \frac{193.30 - 192.90}{193.10} = 20.71$$

HSBC =
midpoint 485.125
spread 0.05
 $((-485.10 + 485.15) / (485.10 + 485.15)/2)$
 $*10^4 = 0.2576$

the cost of an immediate round trip
investment journey

Spreads IV

Interpretation:

- Buying and then instantly selling a unit of stock will cost you 20.71 b.p.
- So we view either buying or selling as costing half the spread (i.e. 10.355 b.p.)
- **Example:** if one was able to buy £10,000 of DOCS stock at the best offer, the spread cost would be $\frac{1}{2} \times 0.002071 \times 10,000 \approx \text{£}10.357$.

Who cares about spreads? Spreads are an excellent measure of liquidity for those trading in small size and those trading infrequently. For a large institution executing a £10 million order, the spread cost is the starting point — **market impact** (the price movement caused by the trade itself) will dwarf the spread cost.

Depth I

There are various definitions for depth but a simple one is as follows;

- **Depth at best:** the average of the total value of stock available at the best bid and the best offer. So if Q_b and Q_a are shares available at the bid and ask we have;

$$\text{Depth} = \frac{1}{2} [Q_a \times P_a + Q_b \times P_b]$$

$500 \times 193.3 + 635 \times 192.9$

So this just tells you roughly how much you can trade, in currency terms, averaged across the best quotes.

This measure is appropriate if one just wants an overview of liquidity and, importantly, has not already decided whether one is a buyer or seller.

Depth II

Comments;

- A trade larger than the quantity at the best quote will necessarily move the market – you will walk up (or down) the order book, paying progressively worse prices.
- Bigger depth → the larger a trade needs to be to instantly move the market.
- **Market impact** is closely related to depth. More depth → less market impact per unit traded.
- Could scale depth by the average daily value traded. This tells you how much of daily volume is currently displayed at best bid and offer.

Depth III

For our HSBA data from earlier on we have;

dispassionate observer perspective of the market

$$\text{Average depth at best} = \frac{1}{2} \times [(4.851 \times 2,056) + (4.8515 \times 2,212)] = \text{£}10,352.587$$

DOCS
500 * 193.3 + 635 * 192.9 = 219,141.5

Comments:

- With price just below £5 per share, depth at both the bid and offer is around £10,000
- So one could immediately buy or sell roughly £10,000 of HSBA stock without moving the market
- Note: often fairly strong asymmetries in separate bid and offer side depth, which gives a useful sentiment signal.

Depth IV

Who cares about depth? Primarily those trading in large size – institutional investors, hedge funds – and those trading repeatedly throughout the day.

- For a retail investor placing a €5,000 order, depth is irrelevant.
- For a portfolio manager executing a €50 million block, it is the binding constraint.

Liquidity, spreads and depth: summary

In a liquid market, we would expect;

- Low spreads: one can trade small amounts cheaply
- High depth (equivalently low market impact): large trades don't cost much more than small trades and don't move markets.

An additional feature we might expect of a liquid market is **resilience**

- In a resilient market, after a large trade drains quantity from a market, that quantity is quickly re-supplied (and at a price close to the original price).

Overall: if we have all three of these things at the same time we can call the market liquid.

- But note, how you trade and how much you trade will influence which aspects of liquidity you care most about.

Overall summary

What we have covered gives the fundamentals of market structure and how we measure market quality. The world is more complex:

- Modern equity markets are fragmented across a wide range of venue types – CLOBs, MTFs, dark pools, SIs, and periodic auctions – with primary exchanges now accounting for less than a third of European trading volume
- A trader's level of patience varies over time, as does the appropriate venue choice
- Routing decisions are not neutral – payment for order flow and varying venue toxicity mean that where your order goes matters
- Liquidity varies over time with stock-specific and market-driven components

In the supplementary materials for this week, we explore liquidity variation across stocks and across the trading day. In Lecture 2, we examine what drives the liquidity we have observed here.